

Title: Zhi-Ecology: An Axiom-Based Framework for Individual Cognition and Civilizational Evolution

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Abstract

This paper proposes "Zhi-Ecology" (智质生态学), a formal theoretical framework designed to unify the explanation of complex system behaviors from individual cognition to civilizational evolution. The framework is built upon five core axioms, from which 24 interlinked theorems are derived. We define key concepts such as "Zhi Ecological Niche" (智质生态位), "Cluster Mind" (集群心智), and "Meta-Cognitive Cluster Intelligence" (元认知集群智能).

The innovation of this framework lies in its integration of individual cognitive dilemmas (e.g., the crisis of meaning) and macroscopic social structural issues within a single explanatory system. As a proof of principle, we demonstrate how applying this framework can guide a general-purpose Artificial Intelligence (AI) to exhibit higher-order cognitive features, such as multi-dimensional perspective-taking and meta-cognitive monitoring. This successfully validates the theory's potency and practical potential.

Our framework not only provides a unified perspective for understanding cognitive and social systems but also demonstrates direct transformative potential and application value in the fields of AI alignment and cognitive enhancement.

Keywords: Zhi-Ecology, Cognitive Relativity, Axiomatic Systems, Complex Systems, Cognitive Science, AI Alignment, Civilizational Evolution